

3.1 Predicates and Quantified Statements I (Homework)

 INSTRUCTOR

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Due Date: THU, SEP 24, 2026 11:59 PM EDT

 **REQUEST EXTENSION**

Current Score: – / 100 POINTS | 0.0 %

Scoring and Assignment Information

QUESTION	1	2	3	4	5
POINTS	– / 20	– / 20	– / 20	– / 20	– / 20

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

1. **[– / 20 Points]**

DETAILS

MY NOTES

When $x = -2$ and $y = 1$, is $Q(x, y)$ true or false?

READ AND COMPLETE THE FOLLOWING.

The hypothesis of $Q(-2, 1)$ is , which is . The conclusion is , which is . Thus $Q(-2, 1)$ is a conditional statement with a hypothesis and a conclusion. So $Q(-2, 1)$ is .

- (b) Give values different from those in part (a) for which $Q(x, y)$ has the same truth value as in part (a).

$$(x, y) = \left(\begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \right)$$

- (c) When $x = 3$ and $y = 8$, is $Q(x, y)$ true or false?

READ AND COMPLETE THE FOLLOWING.

The hypothesis of $Q(3, 8)$ is , which is . The conclusion is , which is . Thus $Q(3, 8)$ is a conditional statement with a hypothesis and a conclusion. So $Q(3, 8)$ is .

- (d) Give values different from those in part (c) for which $Q(x, y)$ has the same truth values as in part (c).

$$(x, y) = \left(\begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \right)$$

Resources

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2. [- / 20 Points]

DETAILS

MY NOTES

PRACTICE ANOTHER

EppDiscMath5 3.1.005.

Find the truth set of each predicate. (If your answer is an interval, enter it using interval notation; otherwise enter it using set-roster notation.)

- (a) Predicate: $\frac{6}{d}$ is an integer, domain: $\mathbf{Z}$

- (b) Predicate: $\frac{6}{d}$ is an integer, domain: $\mathbf{Z}^+$

(c) Predicate: $1 \leq x^2 \leq 4$, domain: $\mathbf{R}$

☐

(d) Predicate: $1 \leq x^2 \leq 4$, domain: $\mathbf{Z}$

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3. [- / 20 Points]

DETAILS

MY NOTES

EppDiscMath5 3.1.010.

Consider the following statement.

$\forall a \in \mathbf{Z}, \frac{(a-1)}{a}$ is not an integer.

(a) Select the correct negation for this statement.

☐

There is an integer a such that $\frac{a-1}{a}$ is not an integer.

☐

For every integer a , $\frac{a-1}{a}$ is an integer.

☐

There is an integer a such that $\frac{a-1}{a}$ is an integer.

☐

For every integer a , $\frac{a-1}{a}$ is not an integer.

(b) Is the given statement true or false? If the statement is true, enter TRUE; if the statement is false, enter a value of a that could be used as part of a counterexample that justifies its falseness.

☐

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4. [- / 20 Points]

DETAILS

MY NOTES

Let D be the set of all students at your school, and let $M(s)$ be " s is a math major," let $C(s)$ be " s is a computer science student," and let $E(s)$ be " s is an engineering student." Express each of the following statements using quantifiers, variables, and the predicates $M(s)$, $C(s)$, and $E(s)$.

(a) There is an engineering student who is a math major.

- ☐ $\exists s \in D$ such that $E(s) \wedge M(s)$
- ☐ $\forall s \in D, E(s) \wedge \sim M(s)$
- ☐ $\forall s \in D, E(s) \wedge M(s)$
- ☐ $\exists s \in D$ such that $\sim E(s) \wedge \sim M(s)$
- ☐ $\exists s \in D$ such that $E(s) \wedge \sim M(s)$
- ☐ $\forall s \in D, \sim E(s) \wedge \sim M(s)$

(b) Every computer science student is an engineering student.

- ☐ $\forall s \in D, \sim C(s) \rightarrow \sim E(s)$
- ☐ $\exists s \in D$ such that $C(s) \rightarrow E(s)$
- ☐ $\exists s \in D$ such that $E(s) \rightarrow C(s)$
- ☐ $\forall s \in D, C(s) \rightarrow E(s)$
- ☐ $\forall s \in D, E(s) \rightarrow C(s)$
- ☐ $\exists s \in D$ such that $\sim C(s) \rightarrow E(s)$

(c) No computer science students are engineering students.

- ☐ $\forall s \in D, \sim E(s) \rightarrow C(s)$
- ☐ $\forall s \in D, C(s) \rightarrow \sim E(s)$
- ☐ $\forall s \in D, \sim C(s) \rightarrow E(s)$
- ☐ $\exists s \in D$ such that $E(s) \rightarrow C(s)$
- ☐ $\exists s \in D$ such that $C(s) \rightarrow \sim E(s)$
- ☐ $\exists s \in D$ such that $\sim C(s) \rightarrow E(s)$

(d) Some computer science students are also math majors.

- ☐ $\forall s \in D C(s) \wedge \sim M(s)$
- ☐ $\exists s \in D$ such that $C(s) \wedge M(s)$
- ☐ $\exists s \in D$ such that $C(s) \wedge \sim M(s)$
- ☐ $\forall s \in D, C(s) \wedge M(s)$
- ☐ $\exists s \in D$ such that $C(s) \wedge M(s) \wedge E(s)$
- ☐ $\forall s \in D, C(s) \rightarrow M(s)$

(e) Some computer science students are engineering students and some are not.

- ☐ $\forall s \in D, C(s) \rightarrow (E(s) \wedge \sim E(s))$
- ☐ $(\exists s \in D \text{ such that } C(s) \wedge E(s)) \wedge (\exists s \in D \text{ such that } C(s) \wedge \sim E(s))$
- ☐ $(\exists s \in D \text{ such that } C(s) \wedge E(s)) \wedge (\exists s \in D \text{ such that } E(s) \wedge \sim E(s))$
- ☐ $(\forall s \in D, C(s) \wedge E(s)) \wedge (\forall s \in D, C(s) \wedge \sim E(s))$
- ☐ $\forall s \in D, (E(s) \rightarrow C(s)) \wedge (\sim E(s) \rightarrow \sim C(s))$
- ☐ $\exists s \in D \text{ such that } C(s) \rightarrow (E(s) \wedge \sim E(s))$

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5. [- / 20 Points]

DETAILS

MY NOTES

PRACTICE ANOTHER

EppDiscMath5 3.1.029.

Let the domain of a variable x be the set of geometric figures in the plane, and let $\text{Square}(x)$ be " x is a square" and $\text{Rect}(x)$ be " x is a rectangle." For each of the statements in (a), (b), and (c), select the answer that expresses the statement without using quantifiers or variables. Then indicate whether the statement is true or false.

(a) $\exists x$ such that $\text{Rect}(x) \wedge \text{Square}(x)$

- ☐ For every geometric figure, if the geometric figure is a square then it is also a rectangle.
- ☐ There are some geometric figures that are rectangles, but are not squares.
- ☐ All geometric figures are both rectangles and squares.
- ☐ There are some geometric figures that are both rectangles and squares.
- ☐ For every geometric figure, if the geometric figure is a rectangle then it is also a square.

☐ All geometric figures are rectangles and not squares.

READ AND COMPLETE THE FOLLOWING.

This statement is .

(b) $\exists x$ such that $\text{Rect}(x) \wedge \sim \text{Square}(x)$

- ☐ For every geometric figure, if the geometric figure is a square then it is also a rectangle.
- ☐ There are some geometric figures that are rectangles, but are not squares.
- ☐ All geometric figures are both rectangles and squares.
- ☐ There are some geometric figures that are both rectangles and squares.
- ☐ For every geometric figure, if the geometric figure is a rectangle then it is also a square.
- ☐ All geometric figures are rectangles and not squares.

READ AND COMPLETE THE FOLLOWING.

This statement is .

(c) $\forall x, \text{Square}(x) \rightarrow \text{Rect}(x)$

- ☐ For every geometric figure, if the geometric figure is a square then it is also a rectangle.
- ☐ There are some geometric figures that are rectangles, but are not squares.
- ☐ All geometric figures are both rectangles and squares.
- ☐ There are some geometric figures that are both rectangles and squares.
- ☐ For every geometric figure, if the geometric figure is a rectangle then it is also a square.
- ☐ All geometric figures are rectangles and not squares.

READ AND COMPLETE THE FOLLOWING.

This statement is .

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